

Noise-Level Alarm



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Noise is a serious environmental problem that affects us in our daily life. There is scientific evidence supporting that noise exposure can cause hearing loss,

hypertension, heart disease, annoyance, sleep disturbance and decreased performance in schools. Sound-level indicators like the one presented here can help address this problem. This sound-level indicator monitors the sound level and indicates through an LED when the level is above the preset value.

Test Points

Test point	Details
TP0	0V, GND
TP1	6V
LED1	Glowing when sound level above threshold
LED2	Glowing when battery connected

Circuit and working

Fig. 1 shows the circuit of noise-level alarm. The circuit is built around three BC550 npn general-purpose transistors (T1 through T3), electret microphone (MIC1), two LEDs (LED1 and LED2)

and a few other components.

The sound is captured by microphone MIC1 and amplified by first-stage high-gain transistor T1. Trimmer potentiometer VR1 is used to adjust the threshold level. The signal is amplified again with second-stage transistor T2. This amplified signal is rectified by diode D1 and the charge is stored in capacitor C10. Diode D1 should preferably be a small-signal Schottky diode such as BAT81, BAT82, BAT83, BAT85 or better. You can use 1N4148 and 1N914 also but the signal from the input should be stronger.

When the voltage across capacitor C10 is high enough, transistor T3 conducts and LED1 glows to indicate that the sound level is higher than the set level. LED2 indicates power supply is available to the circuit.

Transistors T1, T2 and T3 should be high-gain type, such as BC550C, BC109C and BC108C. For powering the circuit, you can use 6V from four AA-size batteries or 6V from a regulated wall adaptor.

Construction and testing

An actual-size, single-side PCB for the noise-level indicator is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in

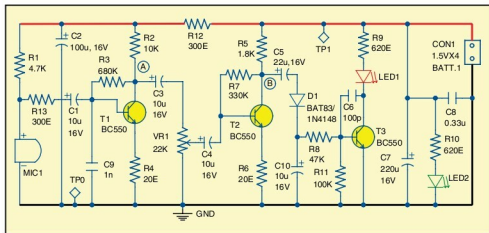


Fig. 1: Circuit of noise-level alarm

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